SAFWAT ALI KHAN

Phone: 571-789-9371 Email: skhan89@gmu.edu

RESEARCH INTEREST

Software Testing, Verification and Validation, Mobile UI Testing, AI Agentic Models, Large Language Models for Software Engineering, Software Accessibility, Automated Program Repair, Trustworthy AI.

EDUCATION

George Mason University *September 2019 - Present*
PhD in Computer Science CGPA: 3.88/4.00
MS in Computer Science *(2025)* CGPA: 3.85/4.00
University of Dhaka *January 2010 - May 2014*
BSc in Computer Science & Engineering

GRADUATE RESEARCH EXPERIENCE

Current projects:

- Building a benchmark for the automated program repair of accessibility defects in software applications (manuscript currently under review).
- Conducting a systematic literature review of developer-facing tools for software accessibility, surveying development guidelines, software-enhancement techniques, targeted impairment populations, input types, experimental methodologies, evaluation metrics, and the SE lifecycle stages and tasks impacted.

Publications:

- "Testing Practices, Challenges, and Developer Perspectives in Open-Source IoT Platforms" - Daniel Rodriguez-Cardenas, **Safwat Ali Khan**, Prianka Mandal, Adwait Nadkarni, Kevin Moran, Denys Poshyvanyk in ICST 2025.
- "Advancing Mobile UI Testing by Learning Screen Usage Semantics" - **Safwat Ali Khan** in ICST 2025.
- "AURORA: Navigating UI Tarpits via Automated Neural Screen Understanding" - **Safwat Ali Khan**, Wenyu Wang, Yiran Ren, Bin Zhu, Jiangfan Shi, Wing Lam, Kevin Moran in ICST 2024.
- "On Using GUI Interaction Data to Improve Text Retrieval-based Bug Localization" - Junayed Mahmud, Nadeeshan De Silva, **Safwat Ali Khan**, Seyed Hooman Mostafavi, S M Hasan Mansur, Oscar Chaparro, Andrian Marcus, Kevin Moran in ICSE 2023.
- "AVGUST: Automating Usage-based Test Generation from Videos of App Executions" - Yixue Zhao, Saghar Talebipour, Kesina Baral, Hyojae Park, Leon Yee, **Safwat Ali Khan**, Yuriy Brun, Nenad Medvidovic, Kevin Moran in ESEC/FSE 2022.

GRADUATE TEACHING EXPERIENCE

Teaching Assistant for:

- Mobile Application Development
- Computer Programming for Engineers
- Intro. to Low-level Programming
- Database Systems
- Data Structures

Responsibilities: Mentored undergraduate and graduate students, graded assignments, and led laboratory sessions. As the teaching assistant for Mobile Application Development, supervised students throughout hands-on lab hours and provided individualized guidance in diagnosing and resolving a broad range of development challenges, from debugging and UI implementation to application design.

RESEARCH TALKS

"Leveraging Machine Learning to bypass Android UI tarpits" - Fraunhofer, USA (2023)
"Automated Exploration of Android Apps" - SE Seminar, George Mason University (2022)

ACADEMIC SERVICE

Journal Reviewer

· IEEE TSE 2026

Co-Reviewer

· FSE 2024, ICSE 2023, SANER 2023, ASE 2022

Volunteer

· ICST 2024

FELLOWSHIP

Provost Scholarship - 2026
Sustainability Summer Research Fellowship - 2024
CS Research Initiation Award - 2020

WORK EXPERIENCE

Software Engineer, QA August 2014 - July 2019

Software Quality Assurance Engineer with experience across manual and automated testing of web and mobile applications. Contributed to impactful projects including systems for underserved populations in Bangladesh, web-based learning management systems for healthcare professionals, and health tracking modules for developmentally disabled individuals.

SKILLS

Programming Languages:	Python, Java, C, C++, JavaScript
Machine Learning APIs:	CLiP, sklearn, tensorflow, pyTorch
Test Automation Tools:	Selenium Webdriver, JMeter, UIAutomator, Google Monkey, APE

MENTORSHIP

· Co-mentor for the Aspired Scientist Summer Internship Program, 2022 and 2020.

EXTRA CURRICULAR ACTIVITIES

· Treasurer for the Bangladeshi Graduate Student Association at George Mason University.

REFERENCES

Dr. Kevin Moran

· Assistant Professor of Computer Science
· University of Central Florida
· Email: kpmoran@ucf.edu

Dr. Wing Lam

· Assistant Professor of Computer Science
· George Mason University
· Email: winglam@gmu.edu